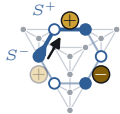


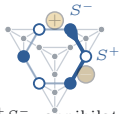
i.a) create

 $S_i^+ S_j^-$: pair splits

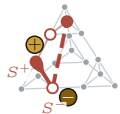
i.b) move

 $S_i^+ S_j^-$: spinon walks

i.c) annihilate

 $S_i^+ S_j^-$: annihilate
 $\mathbf{q}_\gamma = (0, 0, 0)$

i.d) create

 $S_i^+ S_j^-$: pair splitsno walk:
the loop closes
at second order

i.e) annihilate

 $S_i^+ S_j^-$: on an image
 $\mathbf{q}_\gamma = (0, 0, -1)$ 